

Reviewer Pack — Minimal Hodge Decomposition Dimension and Communication Comp...

Entry 6261795535e1 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Decomposition Dimension and Communication Complexity Rank

Entry ID: 6261795535e1 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-08 16:18:43 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Hodge Decomposition) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For any given Boolean function f with n variables, the rank of its communication complexity matrix is upper bounded by the dimension of the minimal Hodge decomposition of the corresponding algebraic variety over the complex numbers, i.e., $\text{rank}(f) \leq \dim(H(f))$, where $H(f)$ is the Hodge decomposition of the algebraic variety associated with f .

Rationale (proposer's reasoning):

Hodge decomposition in algebraic geometry provides a way to study the structure of algebraic varieties. If communication complexity, which measures the amount of information exchanged between parties in a distributed computation, can be related to the geometric properties of these varieties, it may reveal new insights into the complexity of certain computational tasks.

Taxonomy category: ArithmeticHierarchy (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b0c4b99d86a6fe0b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient between the communication complexity matrix ranks and the Hodge decomposition dimensions for all tested Boolean functions with n variables is greater than or equal to 0.8, AND no seed produces a communication complexity matrix rank that exceeds the corresponding Hodge decomposition dimension by more than 3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Hodge Decomposition" AND "communication complexity" AND "matrix rank" - "algebraic geometry" AND "Boolean function" AND "rank bound" - "minimal Hodge decomposition" AND "complex numbers" AND "communication complexity dimension"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.5s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def hodge_dimension(f):
        # Placeholder for Hodge dimension calculation
        # This is a dummy implementation and does not reflect the actual Hodge decomposition.
        return len(f)

    def communication_complexity_matrix_rank(f):
        # Placeholder for communication complexity matrix rank calculation
        # This is a dummy implementation and does not reflect the actual computation.
        return random.randint(0, 10)

    n = 5 # Start with small n and increase to test asymptotic behavior
    instances_tested = 0
    total_rank = 0
    total_dimension = 0

    while True:
        f = generate_boolean_function(n)
        rank = communication_complexity_matrix_rank(f)
        dimension = hodge_dimension(f)

        if rank > dimension + 3:
```

```

    return {
        "metric_name": "communication_complexity_rank",
        "metric_value": rank,
        "instances_tested": instances_tested,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": f"Rank {rank} exceeds dimension + 3 ({dimension})"
    }

    total_rank += rank
    total_dimension += dimension
    instances_tested += 1

    if instances_tested >= 30:
        break

    n += 5

mean_rank = total_rank / instances_tested
mean_dimension = total_dimension / instances_tested
correlation_coefficient = (instances_tested * sum(rank * dim for rank, dim in zip(f, f)) -
                           instances_tested * mean_rank * mean_dimension) / \
    math.sqrt((instances_tested * sum(rank**2 for rank in f) - instances_tested * mean_rank**2) *
              (instances_tested * sum(dim**2 for dim in f) - instances_tested * mean_dimension**2))

    return {
        "metric_name": "communication_complexity_rank",
        "metric_value": correlation_coefficient,
        "instances_tested": instances_tested,
        "n_max": n,
        "conjecture_holds": correlation_coefficient >= 0.8 and all(rank <= dimension + 3 for rank, dimension in zip(f, f)),
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2*i - 1 for i in range(5, 30)]
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif sum(1 for r in results if not r["conjecture_holds"]) / len(results) >= 0.2:
        print(f"RESULT: FALSIFIED counterexample={next(r['counterexample'] for r in results if not r['conjecture_holds'])}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means the pre-registered support condition could not be evaluated. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14354
2	preregistration	ollama_remote	glm4:latest	0	0	9461
3	novelty	ollama_remote	glm4:latest	0	0	8736
4	novelty	ollama_remote	glm4:latest	0	0	9393
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18503
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	35240
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	46677
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	28383
9	judge	ollama_remote	glm4:latest	0	0	51631

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 222379 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/6261795535e1.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/6261795535e1.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/6261795535e1.tar.gz` (if generated)